AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM341
Computer Vision

Midterm Examination (MTA) Makeup
Spring Semester 2024/2025

Date: 22/April/2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 120 Minutes**

Instructions:

- Total Marks: 30
- This exam consists of 3 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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Part – A (5 Marks)

1. Closing is
   a) Dilation followed by Erosion
   b) Erosion followed by Dilation
   c) Dilation
   d) Erosion

2. In Affine transformations
   a) Lines map to lines
   b) Parallel lines remain parallel
   c) Ratios are preserved
   d) All of the above

3. ..... can perform well if the image is scaled differently
   a) SIFT
   b) Harris
   c) SURF
   d) a and c

4. The Prewitt operator is most similar to which of the following filters?
   a) Laplacian filter
   b) Gaussian filter
   c) Canny filter
   d) Sobel filter

5. To map a narrow range of low gray-level input image into a wider range of output levels, we use
   a) Log Intensity Transformation Function
   b) Power-law Intensity Transformation Function
   c) Inverse Log Intensity Transformation Function
   d) Identity Intensity Transformation Function

Part – B (10 Marks)

Question 1: (4 marks).

a) Explain the HSV color model and its components?
a) How does low-level vision contribute to computer vision applications?

Question 2: (3 marks)

a) Give the definition of Laplacian filter
b) The following figures show a 3-bit image of size 5-by-5 image

$$I = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 5 & 3 & 2 \\ 4 & 1 & 2 & 7 & 6 \\ 3 & 5 & 7 & 4 & 1 \\ 2 & 6 & 1 & 3 & 7 \end{pmatrix}$$

Compute the following:

1. The output of the 3 × 3 Laplacian filter = [0 1 0; 1 −4 1; 0 1 0] at value 5 at position (2, 3) indicated in bold.
2. The histogram of the whole image.

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Question 3: (3 marks)

Fill in the blank with the required code:

```
image = cv2.imread('image.jpg')

# Split the image into its Blue, Green, and Red channels
a) _____

#Show the blue channels
b)_____

# Merge the channels back together

c)_____

cv2.imshow('Original Image', image)
cv2.imshow('Merged Image', merged_image)
```

Part – C (15 Marks)

Question 1: (4 marks)

a) Define the Canny Edge Detection algorithm and explain its steps.
b) Write python code to apply the Canny edge detection to an image I.

Question 2: (6 marks)

1. Describe the key steps involved in applying the Harris Corner Detection algorithm to an image.
2. Write a python code to
    1. convert image to grayscale image
    2. detect and dilate corners using Harris
    3. show the results

Question 3: (5 marks)

You are given two grayscale images, Image A and Image B, each with a resolution of 600x600 pixels. Your task is to perform image blending by taking the first 250x250 pixels from Image A and the last 250x250 pixels from Image B.

Write a python code to:

a) Read the two PNG images and extract the first 250x250 pixels of Image A and the last 250x250 pixels of Image B.

b) Blend the extracted regions using a weighted sum, where 30% of the pixel values come from Image A and 70% from Image B.

End of questions

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